Gauri Dubey

FPGA Design Engineer

RESUME_1346

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Total Experience: 5 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 5 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: DSP, VHDL, Intel Quartus, Tcl, UART, FPGA, Xilinx Vivado, Python, ILA, Timing Closure, AXI4, Verilog

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Senior FPGA Design Engineer | QuantaSpring Systems | Jan 2023 - Present

- Reduced FPGA resource utilization by 25% through arithmetic sharing
- Improved maximum operating frequency by 16% after timing-path optimization
- Lowered end-to-end data-path latency by 27% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate FPGA Design Engineer | DeltaPoint Innovations | Jul 2021 - Dec 2022

- Lowered end-to-end data-path latency by 26% using pipelined logic
- Verified 401 protocol corner cases with repeatable simulation suites

PROJECTS

Camera Data Bridge | Xilinx Vivado, DSP, VHDL, Verilog

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Real-Time Signal Filter | Intel Quartus, Python, FPGA, UART

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Shivanta College of Technology, 2021
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Digital Signal Processing Basics (2025)